

LOTHISTORY Business description and related fields:

Executive Summary



Field / Concept	Business meaning
LOT	The unit of work processed through the fab. A lot contains one or more wafers and is the primary WIP and genealogy object.
OPER	A specific process step on a route. It answers: where is the lot in the manufacturing flow?
COMMAND	The transaction or system action performed against the lot. It answers: what action was taken?
HISTCODE	The specific history event or attribute recorded by a command, such as MOVE IN, MOVE OUT, ROUTE, COMMENT, or LOT COMPLETE.
HISTORDER	A field used in lot-history extracts to distinguish or order records that share the same transaction context. The exact database contract should be validated against the authoritative schema.
MASK_LVL	An operation property associated with the process step. Internal documentation confirms operations have a mask level. It is the masking level that operations are associated with. It can also be thought of as layer.
OPERATION_DESC	The human-readable operation description. OPER should be treated as the identifier; descriptions should be interpreted with route and historical context.

1. LOT Lifecycle

What a LOT represents

In NetVision, a LOT is the unit of work processed through the fab. It contains one or more wafers, typically 25, and carries the information needed to manage current WIP position, route, operation, machine assignment, dispatch area, masks, priority, sales-order linkage, attributes, and manufacturing history.

End-to-end business lifecycle

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Stage	Business activity	Typical records or outcomes
1. Define product flow	A part is assigned to a route. The route contains an ordered sequence of runcards and operations.	Approved route and operation sequence.
2. Create the lot	LotCreate establishes the lot with part, route, owner, wafer, mask, history, and initial WIP details. For applicable production lots, raw wafer inventory is decremented in Great Plains.	LTCR command; CR, PT, RT, WF and related history entries.
3. Initial dispatch	After creation, eligible non-test-wafer lots are automatically dispatched to the first operation when required route-operation-machine and recipe information is available.	Initial WIP and dispatch state.
4. Process through route	The lot moves through operations using move-in and move-out activity. NetVision validates resources and captures genealogy, EDC, SPC, comments, status, and location history.	MVIN / MOUT and movement-related history.

5. Handle exceptions	The lot may be held, released, reworked, split, merged, repositioned, rerouted, or adjusted under business rules.	HOLD, REWK, SPLT, LMRG, LRTE, LOPR and related history.
6. Stage for transfer	The completed fabrication lot reaches the TRANSFER operation or staging area and must satisfy transfer validation, including applicable inspection and quality requirements.	Transfer eligibility.
7. Complete and transfer	LotTransfer marks the lot completed, records completion data, removes it from active WIP, and may create a finished-goods inventory transaction in Great Plains.	TRNF command; LC history; closed-to-manufacturing state.
8. Reporting and downstream use	Completed lot information remains available for reporting, labels, quality, traceability, and genealogy.	Reporting record and retained history.

2. Operations and Commands

OPER: the process step

An operation is a specific process step that a lot goes through. At minimum, an operation has an operation number, description, mask level, and dispatch area. Operations are ordered within runcards and routes. A lot is placed at the first route operation when created and progresses through the ordered flow.

COMMAND: the transaction

A command identifies the transaction or applications action that caused the lot-history records to be written. One command can generate multiple HISTCODE rows because a single business transaction may record several effects or attributes.

Command	Plain-Language name	Role in lifecycle
LTCR	Lot Create	Creates a fabrication lot and its initial manufacturing records.
MVIN	Move In	Records entry into processing at the current operation or machine.
MOUT	Move Out	Records completion or exit from the current processing step and may advance the lot.
DISP	Dispatch	Assigns or requests a lot for processing.
HOLD	Hold	Changes or records lot hold status and comments.
SPLT	Lot Split	Creates a child lot or separates wafers from the original lot.
LMRG	Lot Merge	Combines eligible lots or quantities.
REWK	Rework	Routes a full or partial lot through an approved rework path.
LOPR	Change Lot Operation	Adjusts the lot to another operation under business rules.
LRTE	Change Lot Route	Changes the route and records associated operation/route effects.
RPOS	Reposition	Repositions or requests the lot within an operation.
SETP / SETE	Set Production / Engineering Flags	Updates production- or engineering-controlled lot attributes.
RPUR	Repurpose Lot	Recreates scrap material for approved engineering use.

TRNF	Transfer	Completes and closes the lot and performs the final manufacturing-to-inventory handoff when configured.
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Interpretation rule Use OPER to understand process position, COMMAND to understand the initiating transaction, and HISTCODE to understand each detailed effect recorded by that transaction.

3. History Codes and Order

HISTCODE

HISTCODE identifies the specific event, attribute, or state change written to lot history. The command-to-history mapping shows that the same HISTCODE can appear under different commands and that one command commonly produces several history rows.

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Code	History label	Business description
CR	CREATE LOT	Creation of the lot.
IN	MOVE IN	Lot entered processing.
MV	MOVE OUT	Lot exited or completed a processing step.
TX	TRANSITION	Lot is transitioning into move-in.
LA	ARRIVAL	Lot arrived at an operation.
DL	DISPATCH	Lot was dispatched.
SK / SA	SKIP OPER / Auto Skip	Operation was skipped manually or automatically.
RT	ROUTE	Route associated with or changed for the lot.
OP	ADJ OPER	Operation adjustment.
SP	SPLIT LOT	Lot split transaction.

MM / MQ	MERGE LOTS / MERGE QTY	Lot merge and resulting quantity.
RW	REWORK	Lot rework.
R1 / R2	RTE LEAVE / RTE RETURN	Departure from and return to the main route.
HC	HOLD CODE	Lot hold status or reason.
LS	CHG STATUS	Lot status change.
CM / CA	COMMENT / Automation Comment	User or automated explanatory comment.
PT	PART NUM	Lot part number.
MR	MASK/REV	Mask number or revision associated with the lot.
OW	OWNERCODE	Lot owner code.
DD	DUE DATE	Lot due date.
HT	HOT STATUS	Hot-lot flag or priority status.
SO	SALES ORDR	Sales-order association.
QI / QD / QE	Quality Incident / Description / Event	Quality event and supporting description.
SC	ADHOCSCRAP	Ad hoc wafer scrap.
UN	UNSCRAP	Previously scrapped wafers were restored.
LC	LOTCOMPLET	Lot completed during transfer.

HISTORDER

Multiple lines were needed for the history entry. This is the order that they were entered.

4. Other Key Fields

MASK_LVL

This is the Mask level or the layer. Masks are used to define objects/structures. Different layers build different parts of the components.

OPER and OPERATION_DESC

OPER is the operation identifier or number of a step on a route. OPERATION_DESC is the display description is the human readable text associated with an operation in a route. Operation number and description are separate operation properties.

5. How to Read a Lot-History Row

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Field	Example	Interpretation
LOT	6053A0K9	The manufacturing lot whose history is being described.
OPER	50325	The process operation associated with the event.
OPERATION_DESC	LOW_TEMP	The human-readable operation description captured in the extract.
COMMAND	MOUT	The transaction was a Move-Out.
HISTCODE	MV	The detailed history event was MOVE OUT.
HISTORDER	1	Order/differentiation within the transaction context; exact semantics require schema confirmation.

MASK_LVL	026	Operation mask/process-layer value; exact encoding requires data-dictionary confirmation.
COMMENTS	QTY: 25	Additional business detail recorded with the event.